Set - 8

① **AIM**: To design an IOT Network to monitor temperature and humidity sensors.

PROCEDURE:

(i) Open the Cisco Packet Tracer
(ii) Add an IOT Gateway / Home Gateway
(iii) Add temperature and humidity sensors.
(iv) Add IOT devices such as fan, light (or) alarm.
(v) Connect the devices to the IOT Gateway using wireless connectivity.
(vi) Configure the IP address and the IOT settings.
(vii) Monitor the sensor values through the IOT interface
(viii) Control the connected devices based on the sensor readings.

Sample Output:

Temp Sensor: 28°C
Humidity Sensor: 65%

Fan Status: ON
Light Status: ON

Result: Thus, an IOT Network was successfully designed and configured using Packet Tracer.

② **AIM**: To configure Default Gateway and DNS Server settings on 8 systems in Packet Tracer and verify connectivity to a simulated web server.

PROCEDURE:

(i) Open Cisco Packet Tracer.

(ii) Add a switch, 8 PC's and a server.

(iii) Connect all devices using Ethernet Cables.

(iv) Assign IP addresses to the 8 PC's.

(v) Configure the Default Gateway on each PC.

(vi) Configure the DNS server address on each PC.

(vii) Configure the Web server with an IP address and enable HTTP service.

(viii) Use the Command Prompt to test connectivity using ping.

(ix) Open the web browser on a PC and access the web server using its domain name

Sample Output

PC1> ping 192.168.1.10
Reply from 192.168.1.10
Reply from 192.168.1.10
Reply from 192.168.1.10

Packets: Sent = 4, Received = 4, Lost = 0

Result: Thus the Default Gateway and DNS settings were configured successfully and connectivity to the web server was verified.

(3) AIM:

To Capture and analyze SSH packets using wireshark and study their header fields and payload.

PROCEDURE:

(i) Open wireshark and select the required network interface.

(ii) Start Packet Capture.

(iii) Establish an SSH Connection between the client and the server.

(iv) Stop the Capture after SSH communication is completed.

(v) Apply the filter: SSH

(vi) Select a SSH packet from the Captured Packet.

(vii) Analyze the Ethernet, IP, TCP and SSH protocol fields.

(viii) Observe that SSH payload is encrypted and cannot be read as plain text.

Sample Output:

Protocol : SSH
Source IP : 192.168.1.2
Destination IP : 192.1.10
Source Port : ~~22~~ 54321
Destination Port : 22
Transport Protocol : TCP
SSH version : SSH-2
Payload : Encrypted

Result:

Thus, SSH packets were successfully captured and analyzed using Wireshark and their header fields and encrypted payload were studied.

(4) ## AIM:

To implement the Two-Dimensional Parity Check method in C for error detection in a given block of data.

PROCEDURE:

(i) Enter the No. of Rows and columns.
(ii) Enter the binary data.
(iii) Calculate the Parity bit for each row.
(iv) Calculate the Parity bit for each Column.
(v) Display the data with row and column parity.
(vi) Check the received data for parity errors.
(vii) If the Parity doesn't match, an error is detected

Sample Output:

Enter rows and columns : 4 4

Enter data bits :

```
1 0 1 1
0 1 0 1
1 1 0 0
0 0 1 1
```

Data with Row parity :

```
1 0 1 1 1 1
0 1 0 1 1 0
1 1 0 0 0
0 0 1 1 1 0
```

Column Parity : 0 0 0 1

<u>Result</u> : Thus, the two dimensional Parity Check method was successfully implemented in C for error detection.